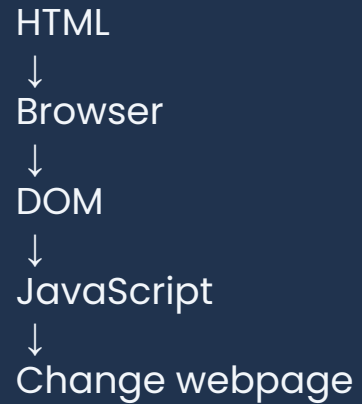


"If JavaScript is a programming language, how does it change something on an HTML webpage?"

The answer is :



1. Selecting and changing an Element

Consider this element
How javascript finds an element

```
<h1 id="title">Hello Students</h1>
```

```
<p class="message">Learn JavaScript</p>
```

```
<button id="btn">Click Me</button>
```

getElementById() - getting element by Id :

```
const heading = document.getElementById("title");  
console.log(heading);
```

```
const heading = document.getElementById("title");
```

```
heading.innerText = "Welcome Students";
```

2. querySelector ()

querySelector() selects the first element that matches a css Selector

Id :
`const heading = document.querySelector("#title");`

Class :
`const message =
document.querySelector(".message");`

Tag :
`const heading = document.querySelector("h1");`

Multiple possibilities
`const button = document.querySelector("button");`
(If there are 5 buttons, it selects only the first button.)

`getElementById()`



ID only

`querySelector()`



CSS selector



first matching element

3. querySelectorAll()

Used when we want to select all matching elements

```
<p class="message">HTML</p>
<p class="message">CSS</p>
<p class="message">JavaScript</p>
const messages = document.querySelectorAll(".message");
```

```
console.log(messages);
It returns a collections of matching elements
```

```
We can use foreach
messages.forEach((message) => {
  message.innerText = "Updated";
});
```

Method

Selects -One element by ID
`getElementById()`

Selects-First matching element
`querySelector()`

Selects-All matching elements
`querySelectorAll()`

4. innerText

Because elements have relationships

```
<h1 id="title">Hello</h1>  
const title = document.getElementById("title");  
  
console.log(title.innerText);  
Output : Hello  
Modify :  
title.innerText = "Welcome Students";
```

5. innerHTML

innerHTML allows us to read or replace the HTML inside an element

```
<div id="content">
  <p>Hello Students</p>
</div>
const content =
document.getElementById("content");

content.innerHTML = "<h2>Welcome!</h2>";
innerText → text
innerHTML → HTML
```

6. value

Value is mainly used in form elements

```
<input id="username" type="text">  
const username =  
document.getElementById("username");  
  
console.log(username.value);  
Changing value :  
username.value = "Student";
```

7. style

Javascript can modify css directly

```
<h1 id="title">Hello</h1>  
const title = document.getElementById("title");  
  
title.style.color = "blue";  
title.style.fontSize = "40px";  
title.style.backgroundColor = "yellow";  
title.style.marginTop = "20px";  
title.style.textAlign = "center";
```

8. classList

Instead of changing individual styles we can add/remove CSS classes

add() methods

remove() methods

```
.highlight {  
  background-color: yellow;  
  color: red;  
}  
<h1 id="title">JavaScript</h1>  
const title = document.getElementById("title");  
  
title.classList.add("highlight");  
  
title.classList.remove("highlight");
```

```
toggle()  
Adds the class if it doesn't exist and removes it  
if it does.  
title.classList.toggle("highlight");  
contains()  
Checks whether the class exists.  
console.log(title.classList.contains("highlight"))  
;  
Returns:  
true  
or  
false
```

9. Example

Student Profile

```
<h1 id="name">Student Name</h1>

<p id="course">Course</p>

<input id="studentInput" type="text">

<button id="updateBtn">Update</button>
.highlight {
  color: blue;
  font-weight: bold;
}
```

```
const name = document.getElementById("name");
const course = document.getElementById("course");
const input =
  document.getElementById("studentInput");
const button =
  document.getElementById("updateBtn");

name.innerText = "Arun";
course.innerText = "JavaScript";

input.value = "New Student";

name.classList.add("highlight");
```

10. Product List Example

```
<ul>
  <li class="product">Laptop</li>
  <li class="product">Mouse</li>
  <li class="product">Keyboard</li>
</ul>
const products =
document.querySelectorAll(".product");

products.forEach((product) => {
  product.classList.add("highlight");
});
```

querySelectorAll()
↓
multiple elements
↓
forEach()
↓
modify each element